M2 Progress Report – Faithful RAG: Evaluating Citation Accuracy and Retrieval Robustness in Domain-Specific Scientific Literature

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1 Progress Summary

All four proposal components run end-to-end on a 37-question gold benchmark of intellectual-property and innovation-policy papers. The retrieval ablation grew from four to sixteen configurations with a new corpus-level SOTA (§2); the faithfulness verifier exposes a 55–61 pp gap between NLI and LLM-judge labels (§3); RAGAS pits chunked RAG against a 1M-token long-context baseline and adds an answer-LLM cost dimension (§4); the Corrective RAG router lifts retrieval correctness from 80% to 90% under a (τ_h, τ_l) sweep (§5).

2 Retrieval Ablation

The 37-question gold set (5 categories incl. multi-hop, byte-validated character spans, 6 adversarial controls q900–q905 that break the single-class Cohen-κ paradox (Cohen, 1960) from the first IAA round) drives a sixteen-configuration ablation. Configurations cross four embedder families (OpenAI `text-embedding-3-small`; BGE-M3 (Chen et al., 2024); E5-large (Wang et al., 2024); ColBERTv2 (Santhanam et al., 2022)), two chunk granularities (coarse ∼2000 chars, fine ∼300 chars), and an optional ZeroEntropy cross-encoder reranker, sharing one `HybridRetriever` code path that reports hit@k, MRR, and nDCG.

config (`coarse_rerank`)	hit@5	hit@10	MRR
OpenAI `text-emb-3-small`	0.946	0.946	0.842
BGE-M3	0.946	0.946	0.842
E5-large	**0.973**	**0.973**	**0.878**
ColBERTv2	0.892	0.919	0.786

Table 1: Best per embedder; full grid in `retrieval_eval/`.

E5-large with the reranker wins every paper-level metric, beating the closed-source OpenAI baseline by +8.6 pp MRR; ColBERTv2 trails by 17 pp MRR. Multi-hop hit@10 falls from 1.000 on coarse to 0.667–0.833 on fine: the chunker is the dominant bottleneck.

3 Faithfulness Verification

The verifier extends the proposal's NLI pipeline (DeBERTa-v3-large-mnli, He et al., 2021) with an LLM-judge head. On 24 answerable gold questions answered closed-book by three generators (GPT-4o-mini, Claude-3.5-haiku, DeepSeek-chat), NLI marks $18/24$ (0.750) faithful at every provider while the LLM judge marks 0.138/0.173/0.198 — a 55–61 pp gap matching a known NLI failure mode (Min et al., 2023; Gao et al., 2023): NLI alone fails as a faithfulness gate.

4 End-to-End RAG vs Long-Context

Two pipelines run through the four RAGAS metrics (Es et al., 2024) on an n=8 stratified subset: chunked RAG (§2 retriever, GPT-4o-mini) and a long-context baseline that loads every cited `document.md` into a 1M-token Gemini 2.5 Flash window, judged by GPT-4o-mini. LC wins every metric but gaps are small (RAG/LC): faithfulness 0.899/0.933, answer_relevancy 0.713/0.789, context_precision 1.000/1.000, context_recall 0.917/1.000 — the open recall gap replicates the chunker bottleneck from §2. OpenRouter cost captures show LC at ∼10× the answer-LLM spend ($0.00383 vs $0.00038/query), reframing the trade-off as accuracy-per-dollar.

5 Corrective RAG and M3 Plan

The CRAG router (Yan et al., 2024) scores top-k retrievals as CORRECT/AMBIGUOUS/INCORRECT and re-queries on AMBIGUOUS. A 9-cell (τ_h, τ_l) sweep on a 10-question subset lifts the 80% baseline correct-retrieval rate to 90% at $(\tau_h$=0.7, τ_l=0.4) for 0.96 s mean latency; τ_h=0.8 drops to 70 % as refinement on already-correct retrievals causes semantic drift. M3 closes the multi-hop, coverage, and LC-recall gaps via a sliding-window chunker, ColBERTv2 domain-adaptation, and a 37-question RAGAS sweep over 16 retrievers + CRAG.

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